CONSTANT VORTICITY BOUNDS, THE BKM INTEGRAL, AND QUADRATIC HISTORY GROWTH: TWO DEFINITIONAL IDENTITIES CHECKED IN LEAN 4

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Abstract. The Beale–Kato–Majda (BKM) criterion says that a smooth solution of the three-dimensional incompressible Euler or Navier–Stokes equations can be continued past time T as long as $\int_0^T \|\omega(t)\|_{L^\infty}\,dt < \infty$. A common heuristic error is to read linear growth of a quadratic history functional as a sign of blow-up. We record, as a worked illustration, the bookkeeping for a vorticity sup-norm that is constant equal to M: the BKM integral over $[0,T]$ is MT, finite, while the quadratic history $\int_0^T M^2\,dt$ equals M^2T and grows linearly in T. In the accompanying Lean 4 formalization both quantities are *defined* by these closed forms, and the two theorems hold by `rfl`. This note does not prove or disprove any statement about solutions of the Navier–Stokes equations, and in particular says nothing about the Fefferman (C) or (D) problems.

1. Background

Let u be a smooth solution of the 3D incompressible Navier–Stokes or Euler equations on $[0,T^*)$ with vorticity $\omega = \nabla \times u$. Beale, Kato and Majda [1] proved that if $T^* < \infty$ is the maximal time of existence then $\int_0^{T^*} \|\omega(t)\|_{L^\infty} dt = \infty$. Consequently, any quantity that stays bounded under a bounded $\|\omega\|_{L^\infty}$ is not, on its own, evidence of singularity.

The simplest illustration is the case $\|\omega(t)\|_{L^\infty} \equiv M$ on $[0,T]$. Then

$$\int_0^T \|\omega\|_{L^\infty}\,dt = MT < \infty, \qquad \int_0^T \|\omega\|_{L^\infty}^2\,dt = M^2T,$$

so the quadratic "history" grows without bound as $T \to \infty$ while the BKM integral is finite for each finite T. These two equalities are elementary calculus; the formalization below does not derive them from integration, but takes the right-hand sides as definitions.

2. Formal content

Definition 2.1. For $M, T \in \mathbb{R}$ set $\mathrm{bkm}(M,T) := M \cdot T$ (Lean: `bkmIntegral`) and $\mathrm{hist}(M,T) := M \cdot M \cdot T$ (Lean: `history`).

Theorem 2.2 (`ClayCD.NotBlowup.bounded_M_keeps_bkm_finite`)**.** *For all $M, T \in \mathbb{R}$, $\mathrm{bkm}(M,T) = M \cdot T$.*

Theorem 2.3 (`ClayCD.NotBlowup.history_can_grow_while_bkm_stays_finite`)**.** *For all $M, T \in \mathbb{R}$, $\mathrm{hist}(M,T) = M \cdot M \cdot T$ and $\mathrm{bkm}(M,T) = M \cdot T$.*

Both proofs are `rfl`: they unfold the definitions. Their mathematical content is precisely the observation, made in the previous section, that under a constant bound the BKM quantity is the real number MT (hence finite) while the history is M^2T. The Lean names describe this intended reading; the formal statements are exactly the equalities above.

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3. Scope

The formalization contains no Navier–Stokes or Euler equations, no vorticity field, no integrals, no function spaces, and no blow-up criterion. It does not formalize the BKM theorem itself. It makes no claim toward the Clay Millennium problem; the repository name records that *no* Clay statement (C) or (D) is certified here.

4. Correspondence between claims and Lean declarations

File `lean/NotBlowup.lean`, namespace `ClayCD.NotBlowup`; Lean 4 v4.30.0-rc1 with Mathlib (for $\mathbb{R}$).

Claim	Lean declaration
Definition of bkm, hist	`bkmIntegral`, `history`
Theorem 2.2	`bounded_M_keeps_bkm_finite`
Theorem 2.3	`history_can_grow_while_bkm_stays_finite`

Appendix A. Verification certificate

The formal development was machine-checked as follows.

- Repository:
  `https://github.com/BenFrohman/Clay-CD-not-certified`
- Verified commit:
  `fa2b0998aaa65550892e1c82ee1727495f333d03`
- Lean toolchain: `leanprover/lean4:v4.30.0-rc1`
- Mathlib: tag `v4.30.0-rc1`, commit (from `lake-manifest.json`)
  `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/Clay-CD-not-certified/actions/runs/41794841523`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  `https://github.com/BenFrohman/Clay-CD-not-certified/archive/fa2b0998aaa65550892e1c82ee1727495f333d03.tar.gz`
  `ff15705c211c81390d26844b281edfbe1def08f3e0938c5b6322363ddeae50fa`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 2 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`ClayCD.NotBlowup.bounded_M_keeps_bkm_finite`	`[propext, Classical.choice, Quot.sound]`
`ClayCD.NotBlowup.history_can_grow_while_bkm_stays_finite`	`[propext, Classical.choice, Quot.sound]`

References

[1] J. T. Beale, T. Kato, A. Majda, *Remarks on the breakdown of smooth solutions for the 3-D Euler equations*, Comm. Math. Phys. **94** (1984), 61–66.

[2] C. L. Fefferman, *Existence and smoothness of the Navier–Stokes equation*, in: The Millennium Prize Problems, Clay Math. Inst. (2006), 57–67.

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